Jef Laga

Pavilion E Office EL.06
Department of Pure Mathematics and Mathematical Statistics
Wilberforce Road
Cambridge, UK

jcsl5@cam.ac.uk
https://www.dpmms.cam.ac.uk/ jcsl5/

Research Interests

Algebraic and arithmetic geometry – specifically: curves and abelian varieties, algebraic cycles, interactions between representation theory and algebraic geometry, arithmetic statistics, rational points

Academic appointments

2023–present	Research Fellow, St John's College, Cambridge
2022–2023	Postdoc, Princeton University
2022	Postdoc under EPSRC award, University of Cambridge

Education

2018-2022	PhD in Mathematics, University of Cambridge, Cambridge, UK Advisor: Jack Thorne
2017-2018	MASt in Mathematics (Part III), University of Cambridge Distinction
20014-2017	Bachelor in Mathematics, Ghent University, Ghent, Belgium Summa Cum Laude Erasmus exchange semester in Université Paris-Diderot, Paris, France

Papers and preprints

When published, papers are listed in order of date of publication.

Preprint	*Polarizations, torsors and theta groups* arXiv:2510.24678
Preprint	*Families of curves in Vinberg representations* with B. Romano arXiv:2508.09607
Preprint	*Kummers, spinors and heights* with J. Thorne arXiv:2507.06865

Preprint *Lower bounds on heights of odd degree points of hyperelliptic curves*
with J. Thorne
arXiv:2507.08652

Preprint *100% of odd hyperelliptic Jacobians have no rational points of small height.*
with J. Thorne
arXiv:2405.10224

2025 *The geometry and arithmetic of bielliptic Picard curves*
with A. Shnidman
Journal of the London Mathematical Society, Vol 112 (2025), Issue 5, e70347

2025 *Vanishing criteria for Ceresa cycles*
with A. Shnidman
Compositio Mathematica, Vol 161 (2025), Issue 11, 3017 - 3043

2025 *Ceresa cycles of bielliptic Picard curves*
with A. Shnidman
Journal für die reine und angewandte Mathematik, Vol 2025, no. 821, 23-51.

2024 *A positive proportion of monic odd-degree hyperelliptic curves of genus $g \geq 4$ have no unexpected quadratic points.*
with A. Swaminathan
International Mathematics Research Notices, Vol 2024, Issue 19, 12857–12866

2024 *Rational torsion points on abelian surfaces with quaternionic multiplication*
with C. Schembri, A. Shnidman and J. Voight
Forum of Mathematics, Sigma, Vol 12 (2024), e92

2024 *Graded Lie algebras, compactified Jacobians and arithmetic statistics*
Journal of the European Mathematical Society, published online

2023 *Arithmetic statistics of Prym surfaces*
Mathematische Annalen, Vol 386, 2023, no.1-2, 247–327

2022 *The average size of the 2-Selmer group of a family of non-hyperelliptic curves of genus 3*
Algebra and Number Theory Vol 16 (2022), no.5, 1161–1212.

Grants and awards

2023 Junior Research Fellowship at St John's College: 4 year research-only position awarded to 4 young researchers across the sciences with over 700 applications each year

2022 EPSRC Award: research associate position funding a 9 month project

2020 Smith-Knight Essay Prize: Grade 1.

2018 Pure Mathematics Prize: Highest Distinction in Pure Mathematics in Part III

2014 International Mathematical Olympiad (honourable mention)

Courses taught

Cambridge	Lie Algebras and their Representations (2024)
Princeton	Calculus II (2022), Topics in Representation Theory (2023)

Designed course content/exam and delivered lectures. Very positive evaluations available on request.

Mentorship

2026 N. Partouche: *Equations defining Kummer varieties of hyperelliptic curves* (four month M1 internship ENS Paris–Saclay)

2026 H. Jakes: *Pure motives* (Part III Essay)

2025 J. Miller: *Counting number fields* (Part III Essay)

2024 H. Kalianaki: *Experiments with canonical heights on elliptic curves* (summer research student through the Cambridge Mathematics Open Internships)

2024 G. Kongarttagarn: *An upper bound for the leading constant in Lang's conjecture* (summer research student through the Cambridge SRIM Programme)

2024 Project assistant in the Arizona Winter School to Ben Moonen's project on modified diagonal cycles (outcome: arXiv:2510.07416)

Other teaching

Organized preparatory workshops in algebraic number theory for incoming Part III students by creating course content on Galois theory and complex analysis, 2019-2021

Supervisions (small group teaching) at Cambridge for Undergraduate Algebraic Geometry (2019), Groups, Rings and Modules (2019) and Linear Algebra (2025)

Teaching Assistant for Lie algebras and their Representations, 2019 and 2024

Mentor for final-year high-school project on cryptography and number theory for the Cambridge Maths School, focused on improving access in STEM (2025-2026)

Elementary number theory and geometry to high school students for the Flemish Mathematical Olympiad every year since 2013

Research visits

Lodha Institute for Mathematical Sciences, Mumbai, special semester on arithmetic statistics (December 2025)

Institute for Advanced Study, Princeton, visiting Wei Ho (June 2025)

Hebrew University, Jerusalem, visiting Ari Shnidman (May 2022)

Selected invited talks

Rational points on varieties of Kodaira dimension 0 *and beyond*, Dryburgh Abbey, Scotland, December 2026

Curves, Abelian Varieties and Related Topics, Barcelona, Spain, July 2026

Recent Developments in Arithmetic Statistics, Lodha Institute, Mumbai, India, December 2025

Rational Points, Schney, Germany, August 2025

Algebraic Points on Curves, ICERM, Providence, US, June 2025

Motives, Mapping class groups and Monodromy, Lanzarote, Spain, February 2025

One day Workshop on Abelian Varieties, Barcelona, Spain, January 2025

Explicit Methods in Number Theory, Oberwolfach, Germany, September 2024

The Ceresa cycle in Arithmetic and Geometry, ICERM, Providence, US, May 2024

Plenary speaker Belgian–Dutch algebraic geometry day, Utrecht, NL, April 2024

More than 25 talks in algebraic geometry and number theory seminars in the UK, Belgium, the Netherlands, France, the US, Canada.

Service

Cambridge	Organized reading groups in Cambridge on Bhargavology, Deligne's purity theorem, Mazur's Eisenstein ideal, Galois deformation theory, automorphy lifting theorems and Arakelov intersection theory.
	Organiser of Number Theory Seminar (2023-2025)
Princeton	Organised the arithmetic statistics seminar.
Refereeing	Referee and quick opinions for Advances in Mathematics, Algebra and Number Theory, Annals of Math, Crelle, Epiga, Experimental Mathematics, Forum Math Pi, IJNT, IMRN, JEMS, Journal of Number Theory, Mathematische Annalen, Mathscinet, Transactions of AMS, etcetera.
Outreach	'Classifying Pythagorean triples: where arithmetic meets geometry'. (Lecture for high school students, St John's College, August 2024), 'Dynkin diagrams and the unity of mathematics' (Lecture for summer research students, August 2025), 'What is maths research all about?' (Lecture for high-school students of Cambridge Maths School, February 2026)
	Invited speaker at Ghent University Honours Programme, with the topic: 'Is Mathematics Science?' (in Dutch), November 2020, December 2024, November 2025.
Other	Served as a jury member of the Flemish Mathematical Olympiad (2014-2016)

References

Jack Thorne (PhD supervisor), University of Cambridge, `jat58@cam.ac.uk`

Manjul Bhargava (Postdoc mentor), Princeton University, `bhargava@math.princeton.edu`

Bjorn Poonen, Massachusetts Institute of Technology, `poonen@math.mit.edu`